



Network Interface Identification(NMAP)

Name: FIREWALL FAIRY

Date: 7August, 2026

Objective

To identify the IP address and network configuration of the attacking machine before beginning reconnaissance.

Command

```
ifconfig
```

The `ifconfig` command was executed to determine the network configuration of the Kali Linux system. The output showed that the active network interface was **eth0** with the

IPv4 address **192.168.237.128/24** and MAC address **00:0c:29:ce:d7:79**.

This information established the attacker's position within the local network and confirmed that the system was connected to the **192.168.237.0/24** subnet.

2. Host Discovery (Ping Sweep)

Objective

To identify live hosts on the local network.

Command

```
nmap -sn 192.168.237.128/24
```

A ping sweep was conducted using Nmap's `-sn` option to discover active hosts without performing port scanning. The scan identified two live systems within the subnet:

- 192.168.237.2
- 192.168.237.128 (Kali attacker)

This confirmed that only two reachable hosts were present on the network during the assessment

3. Default TCP Port Scan

Objective

To identify commonly used open TCP ports on the target machine.

Command

```
nmap 192.168.237.128
```

A standard TCP port scan was performed against the target. Nmap reported that **996 ports were closed**, while four services were accessible:

Port	Service
22	SSH
53	DNS
80	HTTP
443	HTTPS

The scan indicated that the target primarily hosted secure remote access, DNS, and web services.

4. Full TCP Port Scan

Objective

To discover all open TCP ports.

Command

```
nmap -p- 192.168.237.128
```

A complete TCP scan covering all **65,535 ports** was performed. The scan confirmed that only the following ports were open:

- 22/tcp
- 53/tcp
- 80/tcp
- 443/tcp

No hidden or additional TCP services were identified beyond those detected in the default scan

5. Network-wide Full Port Scan

Objective

To enumerate services across all live hosts in the subnet.

Command

```
nmap -p- 192.168.237.128/24
```

A subnet-wide full TCP scan was executed. The gateway (**192.168.237.2**) exposed only the DNS service on port 53, whereas the Kali machine exposed SSH, DNS, HTTP, and HTTPS services. This provided a complete inventory of accessible TCP services within the local network.

6. Service Version Detection

Objective

To determine the versions of running network services.

Command

```
nmap -sV 192.168.237.128
```

Service version detection identified the following software:

Port	Service	Version
22	SSH	OpenSSH 9.7p1
53	DNS	ISC BIND 9.20.9
80	HTTP	Apache HTTP Server 2.4.54
443	HTTPS	Apache HTTP Server 2.4.54

These version details are useful for vulnerability assessment and exploit research.

7. Vulnerability Search Using Searchsploit

Objective

To identify publicly known exploits associated with detected services.

Commands

```
searchsploit OpenSSH9.7p1  
searchsploit ISC BIND9.20.9  
searchsploit Apache httpd2.4.54
```

The detected software versions were searched against the Exploit Database using **Searchsploit**. No publicly available exploits were found for the identified versions of OpenSSH, ISC BIND, or Apache HTTP Server. This suggested that these services were either up-to-date or had no known exploits indexed in Exploit-DB at the time of testing.

8. Operating System Detection

Objective

To identify the operating system of the target.

Command

```
sudo nmap -O 192.168.237.128
```

Initially, the OS detection scan failed because Nmap requires root privileges for TCP/IP fingerprinting. After switching to the root user, the scan completed successfully and identified the target as a **Linux-based operating system**, estimating the kernel version as **Linux 2.6.32**.

9. Linux Kernel Exploit Enumeration

Objective

To search for exploits targeting the detected Linux kernel.

Command

```
searchsploit Linux2.6.32
```

Searchsploit was used to enumerate exploits related to Linux Kernel 2.6.32. The tool returned numerous historical privilege escalation and local exploitation references, including several kernel vulnerabilities. These findings indicate that older Linux kernels may contain publicly documented vulnerabilities; however, further verification is necessary before concluding that any specific exploit is applicable to the scanned system.

10. Nmap Default Script Scan

Objective

To gather additional information about exposed services using Nmap's default NSE scripts.

Command

```
nmap -sC 192.168.237.128
```

Nmap's default scripting engine was executed to collect additional service information. The scan revealed:

- SSH host fingerprints (ECDSA and ED25519)
- BIND DNS version information
- Apache default webpage
- Self-signed SSL certificate with common name **root**
- TLS configuration details

The results confirmed that the web server was using the default Apache configuration and a self-signed certificate, which is commonly observed in testing

or development environments